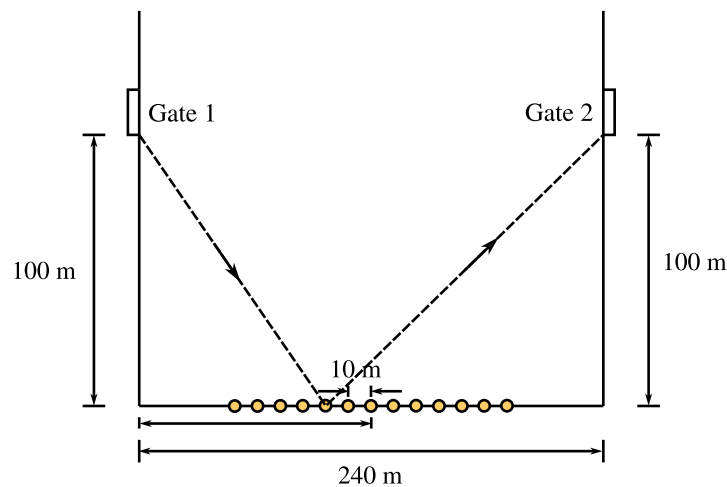


Worksheet: Optimal running paths

Task 1

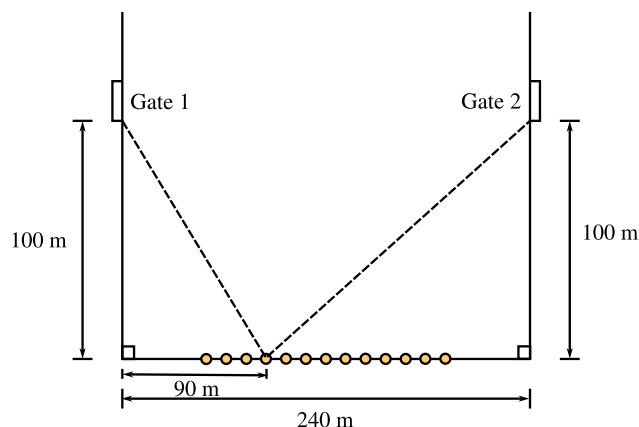
Scenario: a marathon runner runs along some road and enters a rectangular field through a gate (Gate 1) and leaves through another 240m away (Gate 2). The runner has the option to grab a water bottle on 13 tables situated on a line on the southern edge 100 m away from both gates:



The 13 tables are at 60, 70, ... , 180 m away from the west edge along the southern edge of the field as shown in the picture.

1. Pick a table. Write down expressions of the total distance of the runner's path in the field as he enters Gate 1, runs toward your choice of table then leaves through Gate 2, always traveling along straight lines as shown in the picture. Evaluate the expression to compute the total distance.

For instance, we pick table #4 at 90m away from the east edge:



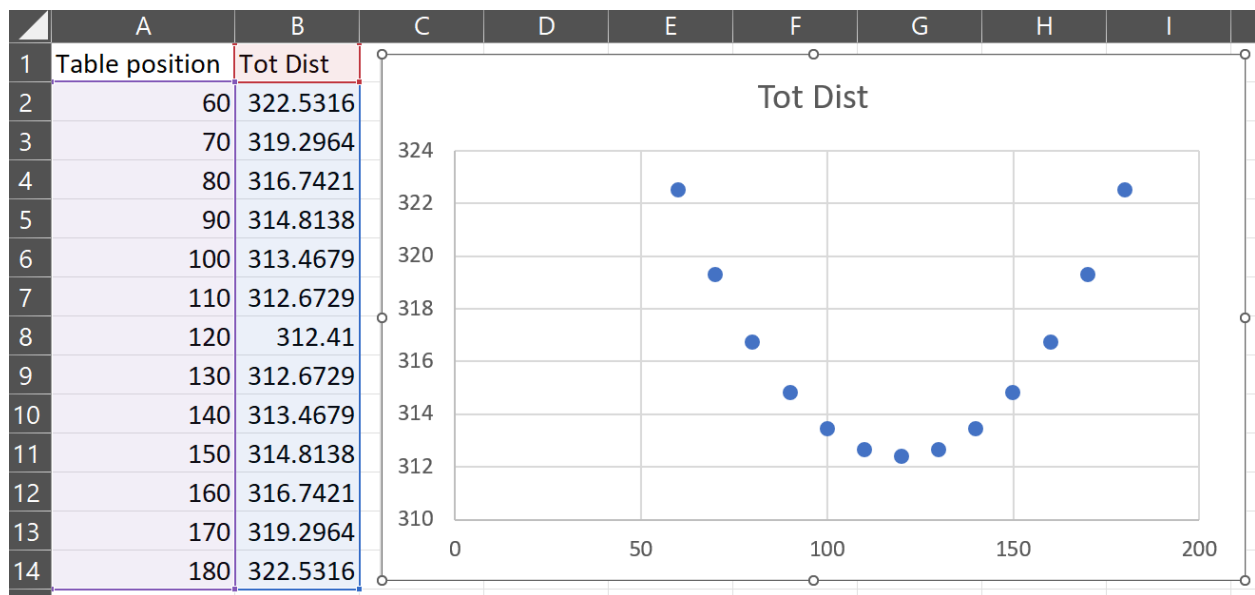
We have: total distance $D = \sqrt{100^2 + 90^2} + \sqrt{100^2 + (240 - 90)^2} \approx 314.81$ m.

- Do so for all 13 tables and fill in the numerical values to nearest centimetre. What are the changes you need to make to your expressions?

Table at...	60m	70m	80m	90m	100m	110m	120m
Tot. Dist.	322.53m	319.30m	316.74m	314.81m	313.47m	312.67m	312.41m
Table at...	130m	140m	150m	160m	170m	180m	
Tot. Dist.	312.67m	313.47m	314.81m	316.74m	319.30m	322.53m	

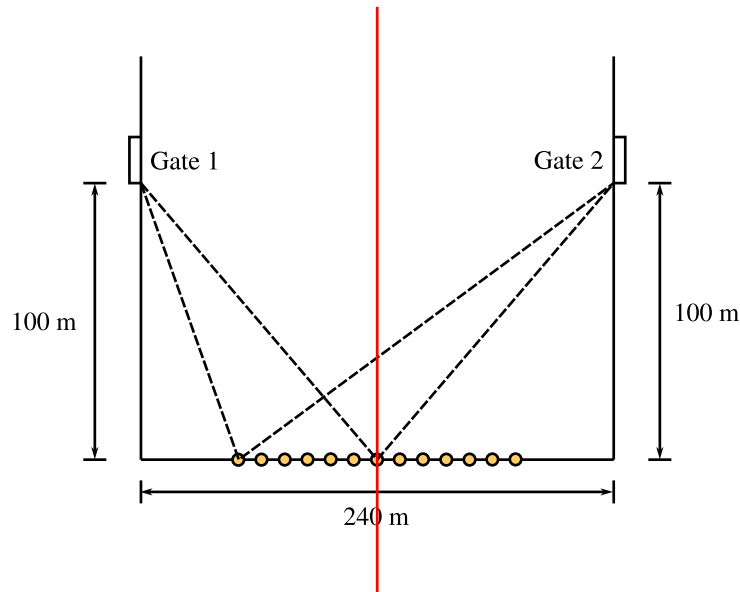
- Graph the data points (total distance vs table choice) using Excel. Which table should the runner choose to run the least distance and save energy for the rest of the marathon?

The optimal choice is the table in the middle, at 120m away from both the east and west edge.



- (Extension) Is there a way you can arrive at this conclusion by computing the total distance for only 7 tables instead of 13?

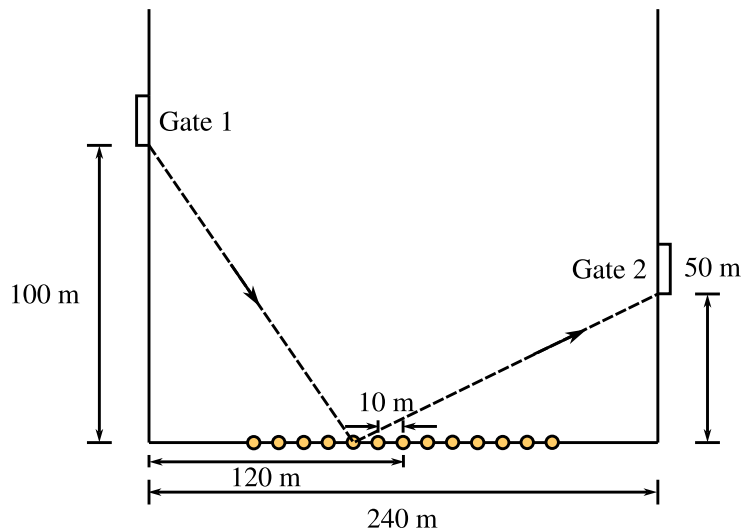
Note: the geometry of the scenario is symmetric across the vertical red line as shown here:



Only tables at 60, 70, ..., 120 m away from the west edge needed to be considered as the right half will give identical values.

Task 2

Scenario: as in Task 1, south edge length is still 240m, but this time Gate 2 is now at 50 m away from the south edge not 100m. The location of Gate 1 is unchanged as in the picture below.



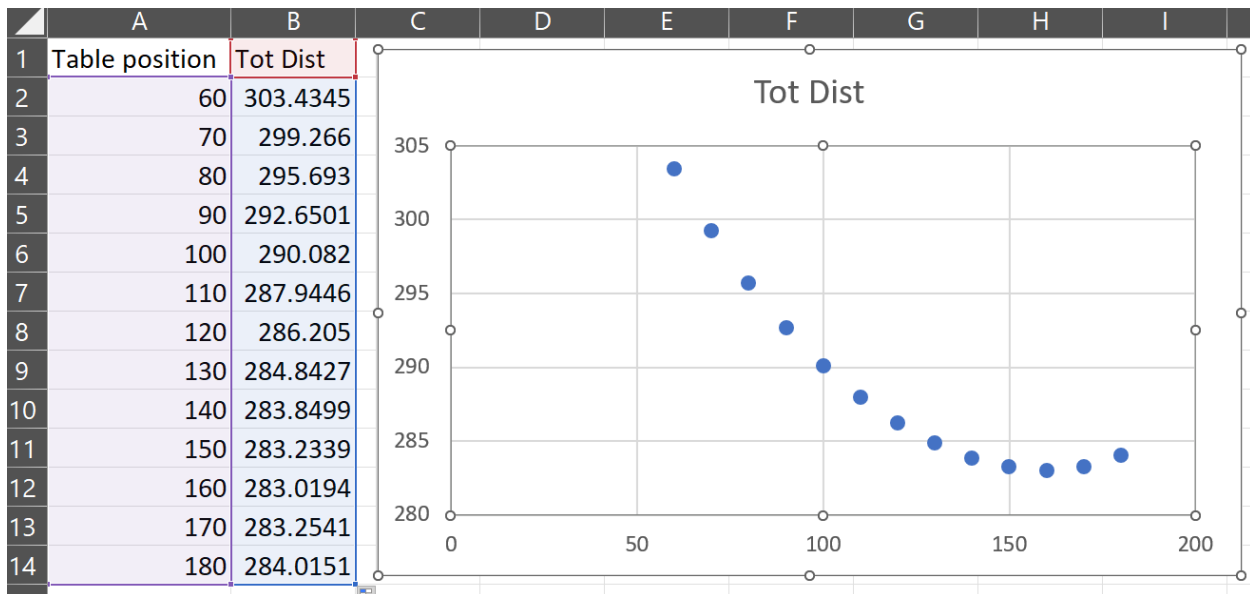
1. Before doing any calculation, write down your guess for which table might be the best choice and how/if you think it differs from the case for Task 1.

This is an open question and it does not have a definitive answer. One possible route is this. In the case for Task 1, the middle table is the best choice. If we change the distance 50 m to

0 m, then the east-most table will be the best choice. We can guess that the trend is that the optimal table will move east-ward.

2. Repeat the calculation in Task 1 for Task 2: compute total distance for the running path for scenario in Task 2 and fill in the table below, then graph using Excel. Which table is the optimal choice this time? Does it confirm or disagree with your guess?

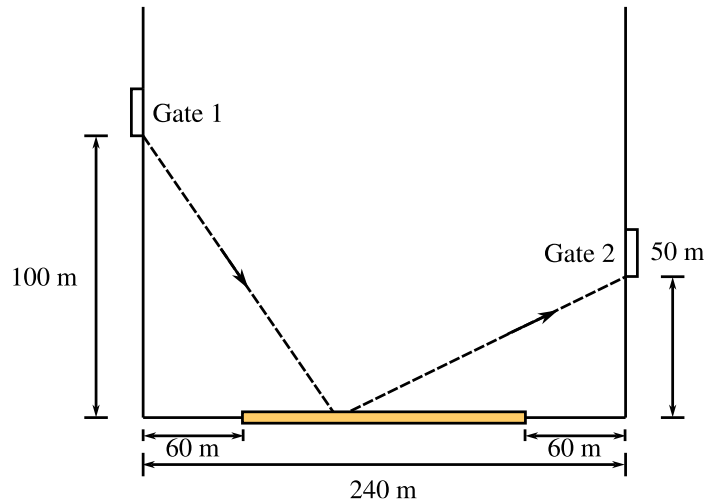
Table at...	60m	70m	80m	90m	100m	110m	120m
Tot. Dist.	303.43m	299.27m	295.69m	292.65m	290.09m	287.94m	286.2m
Table at...	130m	140m	150m	160m	170m	180m	
Tot. Dist.	284.84m	283.85m	283.23m	283.02m	283.25m	284.02m	



The optimal table is at 160m away from the west edge. Its position indeed moved east-ward.

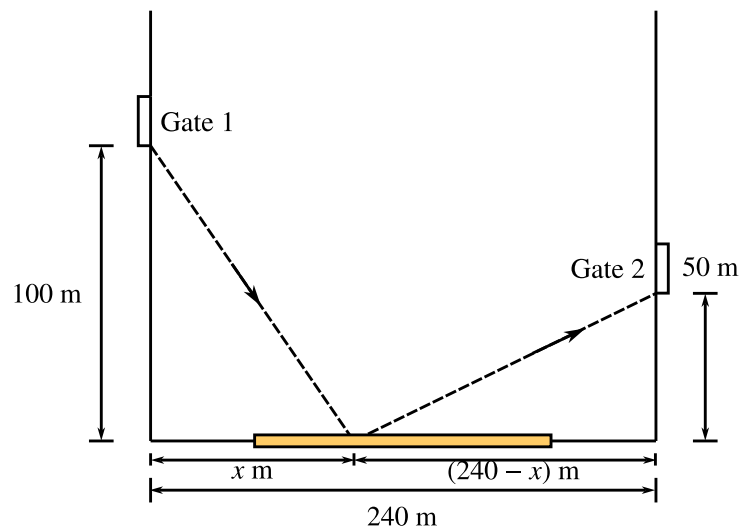
Task 3

Scenario: the field and location of Gates 1 and 2 are identical to Task 2, but this time instead of 13 separate tables, there are a number of tables connected together to form a single long surface on which there are enough water bottles for the runners to grab. One end of the long table is 60 m away from the west edge, the other end is 60 m away from the east edge. No matter where the runner reaches the long table, he/she can grab a water and continue the marathon as shown below:



1. The total distance of the runner now varies continuously with different choices of the location on the long table where the runner chooses to grab the bottle. Set up an independent variable and write down the expression of the total distance in terms of this independent variable.

We may set x to be the distance from the location the runner reaches the long table to the south-western corner of the field as shown here:



Total distance as a function of x : $D(x) = \sqrt{100^2 + x^2} + \sqrt{50^2 + (240 - x)^2}$

2. Find the optimal location which allows the runner to run the minimal distance between the two gates and grab a bottle from the long table. What operation do you need to perform to the function in order to find this optimal value of your independent variable? Why?

The operation to the function is to find critical point (and then verify it is a minimum). We may solve $D'(x) = 0$ for x to find critical points. We have

$$D'(x) = \frac{x}{\sqrt{100^2 + x^2}} + \frac{-(240 - x)}{\sqrt{50^2 + (240 - x)^2}}$$

So the equation reads

$$\frac{x}{\sqrt{100^2 + x^2}} + \frac{-(240 - x)}{\sqrt{50^2 + (240 - x)^2}} = 0$$

From here on, there are several different routes:

- (1) Brute force this solution. This is doable, but cumbersome.
- (2) Hold off until we have expression in Task 4 before computing. The advantage is that we will no longer be tempted to evaluate things like 50^2 and expand $(240 - x)^2$

We will choose the second option here.

3. (Extension) Compare your optimal position to the position of the optimal table in Task 2. Is there any difference? If so, how do you make sense of this difference?

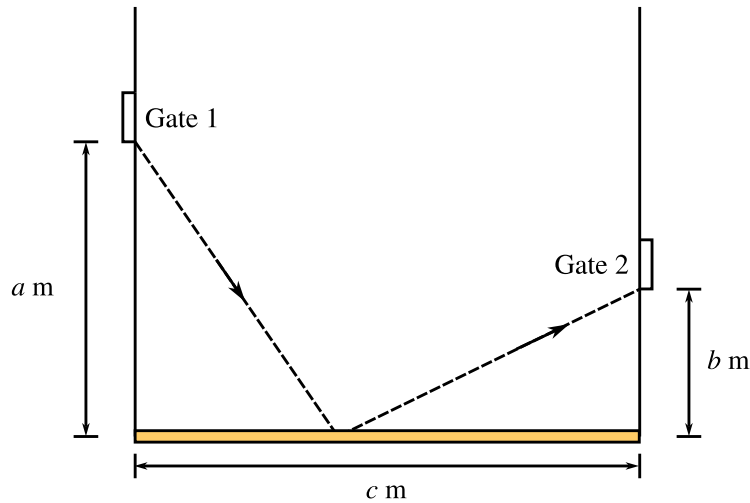
If we borrow the formula from Task 4, the optimal position should be at

$$x = \frac{ac}{a + b} = \frac{100 \times 240}{100 + 50} = 160$$

This gives precisely the position at 160m. However, if we change the location of tables, it may not be the case that our optimal table in the discrete setting is the optimal position in the continuous setting. There will be a difference between the two optimal positions bounded by the distance between two tables in the discrete setting.

Task 4

Scenario similar to Task 3, but this time the table is as long as the southern edge. No actual number is given for the dimensions in this picture below, instead we view a , b , c as known constants.



1. Set up an expression for the total distance and an independent variable to determine the location on the long table where the runner grabs water bottle.
2. Find a simplified formula that provides the optimal position on the long table to minimise runner total distance on the field for any positive number choices for a, b, and c.
3. (Extension) Is there a way we can arrive at this formula without using calculus?

Similar to Task 3, the formula is given by

$$D(x) = \sqrt{x^2 + a^2} + \sqrt{b^2 + (c - x)^2}$$

We may compute the derivative as

$$D'(x) = \frac{x}{\sqrt{x^2 + a^2}} + \frac{-(c - x)}{\sqrt{b^2 + (c - x)^2}}$$

To find critical point, set $D'(x)=0$, or

$$\frac{x}{\sqrt{x^2 + a^2}} + \frac{-(c - x)}{\sqrt{b^2 + (c - x)^2}} = 0$$

From this point there are two routes: (1) purely algebraic method (2) algebraic + geometric arguments.

- (1) Since the square roots each came from Pythagorean theorem, they each represent a positive length, we can multiply through to get

$$x \sqrt{b^2 + (c - x)^2} = (c - x) \sqrt{x^2 + a^2}$$

This still looks difficult, but we may square both sides and get

$$x^2(b^2 + (c - x)^2) = (c - x)^2(x^2 + a^2)$$

Followed by expansion and simplification, we get a quadratic equation in x:

$$(a^2 - b^2)x^2 - 2a^2cx + a^2c^2 = 0$$

Using quadratic formula and simplifying, there are 2 roots

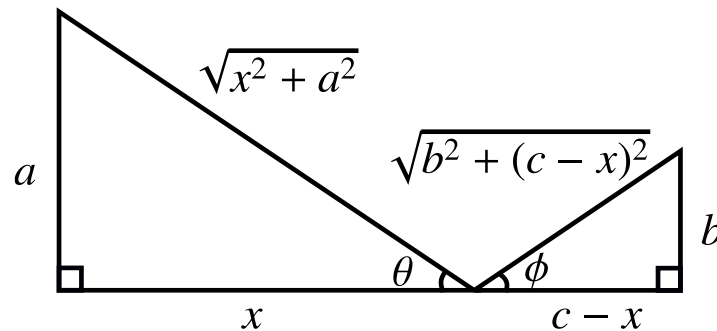
$$x = \frac{ac}{a \pm b}$$

However, not both root correspond to valid critical points. Note that the optimal position will always be positive, even when $a < b$, but in that case the root $ac/(a - b)$ becomes negative. The root to choose is therefore $x = ac/(a + b)$. We may then verify that it is indeed a minimum by either computing $D''(x)$ at this value or other arguments.

(2) If we pause for a second at this equation

$$\frac{x}{\sqrt{x^2 + a^2}} = \frac{(c - x)}{\sqrt{b^2 + (c - x)^2}}$$

And try to apply Pythagorean theorem in reverse to read it geometrically, what it says is:

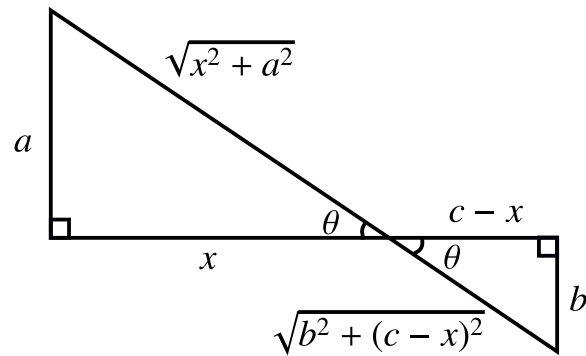


That $\cos \theta = \cos \phi$. But both of the angles are acute, so this is the same as saying $\theta = \phi$. We note that an equivalent but simpler way of expression this equality of angles is via cotangents: $\cot \theta = \cot \phi$, or in other words

$$\frac{x}{a} = \frac{c - x}{b}$$

This is now a linear equation which easily gives $x = \frac{ac}{a+b}$.

As an extension, we can further note that we can reflect either right-angle triangle in the above picture across the southern edge and get



In other words, the optimization problem becomes this: the runner will run from Gate 1 to reflected Gate 2 while moving east and south. Before and after reaching the southern edge, he/she runs in straight lines. The optimal choice is now clearly the straight line, which is the shortest amongst all piecewise-straight paths.

